UNIT 12

ENVIRONMENTAL CHEMISTRY

ENVIRONMENTAL CHEMISTRY

Definition: "It is the branch of chemistry which deals with the chemicals and other pollutants in the environment resulting directly or indirectly from human activities."

Components:

Atmosphere	The layer of gases which surrounds the earth.
Hydrosphere	Concerned with all water bodies i.e. ocean, rivers, streams, lakes & glaciers.
Lithosphere	Concerned with hard and rigid rocky earth crust.
Biosphere	Area on earth which supports life i.e. air, lakes etc.

Atmosphere:

The layer of air surrounding the earth at a height of around 1600 km and it consists of many gases is known as atmosphere.

Composition of Atmosphere:

The atmosphere is composed mainly of nitrogen, oxygen, and other trace gases. It also contains water vapor and suspended particles.

Percentages of gases in atmosphere:

Gas	Percentage in Air
Nitrogen (N_2)	~78%
Oxygen (O_2)	~21%
Argon (Ar)	~0.93%
Carbon Dioxide (CO_2)	~0.04%
Neon (Ne)	Trace Amount
Helium (He)	Trace Amount
Methane (Ch_4)	Trace Amount
Others	Trace Amount

Smog: It is the combination of smoke and fog i.e. 'sm' from smoke & 'og' from fog.

There are two types of smog:

i. Reducing Smog (industrial smog).

ii. Oxidizing Smog (Photochemical Smog).

It is a yellowish brownish grey haze which is formed in the presence of water droplets and chemical reactions of pollutants in the air.

It has unpleasant odour because of its gaseous components. The main reactants of photochemical smog are nitric oxide (NO) and unburnt hydrocarbons.

The yellow colour in photochemical smog is due to the presence of nitrogen dioxide.

Layers of Atmosphere:

In the atmosphere, there are five distinct layers, each with unique characteristics and importance.

1. Troposphere: The layer closest to the Earth's surface where weather occurs. Troposphere constitutes the major portion of the atmosphere.

2. Stratosphere: Contains the ozone layer that absorbs harmful ultraviolet radiation. Also called ozonosphere as ozone layer is found here.

3. Mesosphere: The coldest layer and where meteoroids burn up.

4. Thermosphere: The layer where the International Space Station orbits and where temperature increases with altitude.

5. Exosphere: The outermost layer, merging with space, and contains a thin dispersion of gases.

Properties	Troposphere (Tropo- means "turning" or "changing")	Stratosphere (Stratos-means "layer" or "spread out"	Mesosphere (Meso- means "middle")	Thermosphere (Thermo-means "heat") • Ionosphere • Exosphere
Altitude (km)	0-12	12-50	50-80	Above 80
Temperature range (°C)	15 to -56	-56 to -5	-5 to -93	-93 to 1800

Distinguish between Troposphere and Stratosphere:

TROPOSPHERE	STRATOSPHERE OR OZONOSPHERE
1. Closest to the Earth's surface.	Located above the Troposphere.
2. Where weather phenomena occur.	Generally free of weather phenomena.
3. Temperature decreases with altitude.	Temperature increases with altitude.
4. Contains most of the Earth's atmospheric mass.	Contains the ozone layer, which absorbs UV radiation.
5. Higher water vapor content.	Lower water vapor content.
6. Lower concentration of ozone. (0.1 ppm)	Higher concentration of ozone. (10ppm)

Pollutants:

Pollutants are substances or agents that contaminate the environment and can cause harm to living organisms and ecosystems.

Types of Pollutants:

1. Air pollutants: This is caused by the addition of undesirable substances into the atmosphere. It is of the following two types:
a) Tropospheric pollution: It is caused by gaseous air pollutants like SO_2, NO_2, CO_2, or H_2S.
b) Stratospheric pollution: The damage to the ozone layer by the action of compounds like nitric acid and chloroform carbons constitute this particular type of pollution.

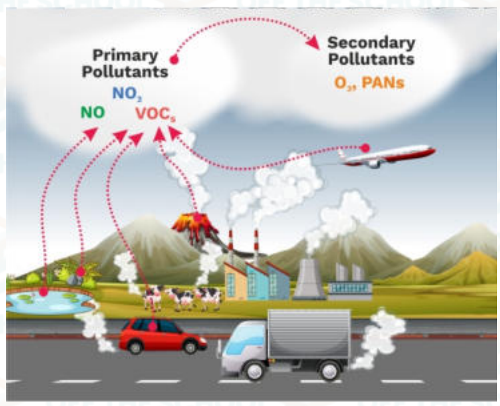
2. Water pollutants	**3.** Soil pollutants	**4.** Thermal pollutants
5. Radioactive pollutants	**6.** Noise pollutants	**7.** Light pollutants

Major Air Pollutants:

There are two types of pollutants:

1) Primary pollutants: that are directly emitted into the atmosphere as a result of human activities or natural processes (SO_2, SO_3, CO, NH_3, Hydrocarbon etc.)

2) Secondary pollutants: which are formed by the primary pollutants (H_2SO_4, H_2CO_3, Hcl, Ozone etc)




Vehicles' smoke


Industrial smoke

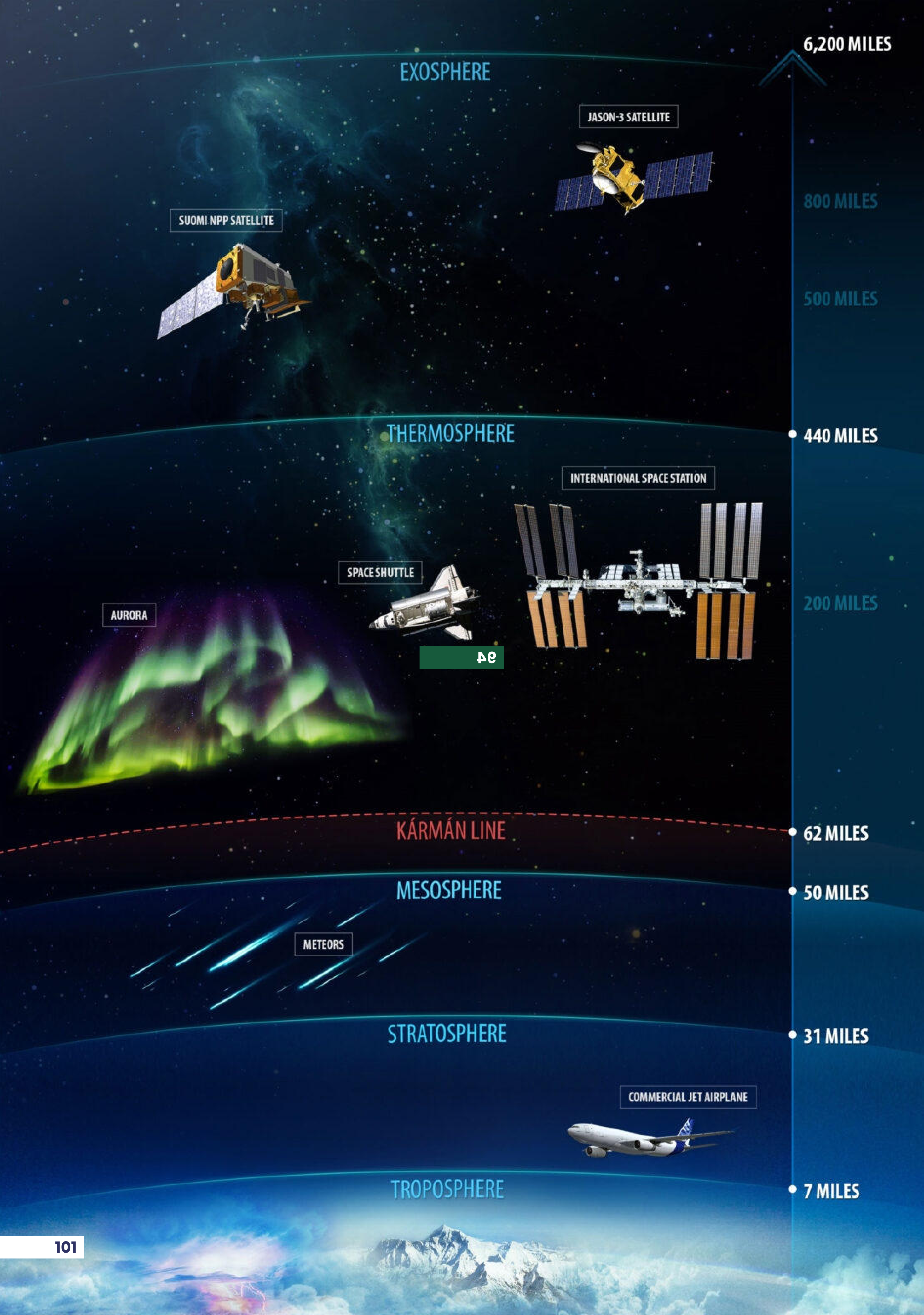

Rotting vegetation


Smoke emitted from thermal power station


CFCs in aerosol spray


Volcanic eruption

Natural sources of air pollutants


6,200 MILES
EXOSPHERE
JASON-3 SATELLITE
800 MILES
SUOMI NPP SATELLITE
500 MILES
THERMOSPHERE
440 MILES
INTERNATIONAL SPACE STATION
SPACE SHUTTLE
AURORA
200 MILES
KÁRMÁN LINE
62 MILES
MESOSPHERE
50 MILES
METEORS
STRATOSPHERE
31 MILES
COMMERCIAL JET AIRPLANE
TROPOSPHERE
7 MILES

- **Atmospheric Pollution**
  - **Tropospheric Pollution**
    - **Gaseous air Pollutants**
      - Oxides of Sulphur
      - Hydrocarbons
      - Oxides of Nitrogen
      - Oxides of Carbon (CO & CO_2)
    - **Particulate Pollutants**
      - Viable Particulates
      - Non-Viable Particulates
        - Smoke
        - Dust
        - Mist
        - Fumes

Major Air Pollutants and Their Impacts

Pollutant	Sources	Environmental Risks	Human Health Risks
Carbon Monoxide (CO)	Incomplete combustion of fossil fuels, vehicle emissions	Contributes to global warming and reduces air quality	Headache, brain damage and affects the cardiovascular system.
Nitrogen Oxides (Nox)	Combustion of fossil fuels, industrial processes	Contributes to smog and acid rain, harms ecosystems	Respiratory irritant, leads to respiratory problems.
Sulfur Dioxide (So_2)	Burning of fossil fuels, industrial activities	Causes acid rain, damages vegetation and water bodies	Irritates respiratory system, lung diseases (Bronchitis, emphysema & lung cancer).
Ozone (O_3)	Formed by reactions of NOx and volatile organic compounds (VOCs) in sunlight	Damages crops and forests, reduces photosynthesis	Respiratory irritant, causes breathing difficulties.
Particulate Matter (PM)	Combustion processes, industrial emissions	Reduces visibility, contributes to haze and acid rain.	Respiratory and cardiovascular problems.
Lead (Pb)	Exhaust fumes from motor vehicles.	Contaminates soil and water, harms wildlife	Affects the nervous system, developmental issues.

Acid Rain and its Effects

Acid Rain:

"when the pH of rain water drops below to 5.6 is called acid rain." Or It can be defined as: "rain water having pH less than 05."

Acid rain is a type of environmental pollution that occurs when acidic substances, such as sulfur dioxide and nitrogen oxides, react with atmospheric moisture and fall to the ground in the form of rain, snow, or dry particles.

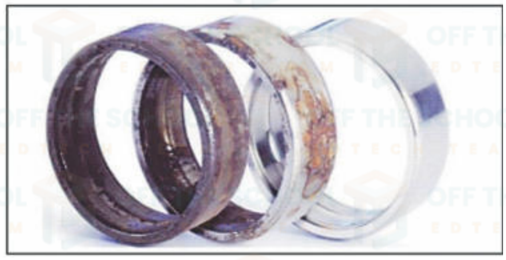
List out acids present in Acid Rain:

1. Sulfuric acid (H_2SO_4)
2. Nitric acid (HNO_3)
3. Carbonic acid (H_2CO_3)

Fish killed by acid rain water

Effect of acid rain on trees

Stonework erosion caused by acid rain



Corrosion of metals caused by acid rain

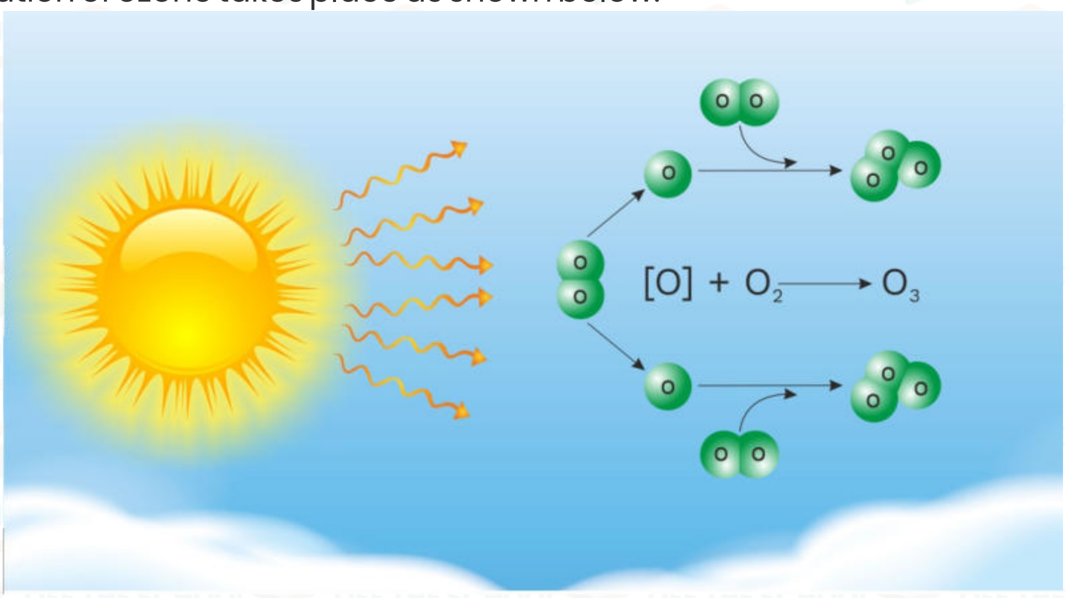
Effects of acid rain

OZONE DEPLETION AND ITS EFFECTS

Ozone Formation:

Ozone formation occurs naturally in the Earth's atmosphere through a process involving ultraviolet (UV) radiation from the sun. UV radiation splits oxygen molecules (O_2) into individual oxygen atoms (O). These oxygen atoms then combine with other oxygen molecules to form ozone (O_3).

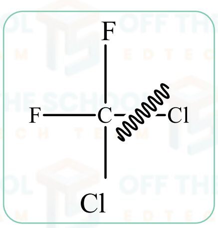
Formation of ozone takes place as shown below:



Ozone Depletion:

Ozone depletion refers to the destruction of the ozone layer in the stratosphere due to human-made chemicals, such as Chlorofluoro carbons (CF_2Cl_2) called freons, absorb the ultraviolet radiations and get photolyzed to liberate free chlorine atoms.

The free chlorine atoms catalyze the decomposition of ozone resulting in the depletion of ozone layer.



$$CF_2Cl_2 \xrightarrow{hV} CF_2Cl + Cl^0$$

$$CFCl_3 \xrightarrow{hV} CFCl_2 + Cl^0$$

$$Cl^0 + O_3 \rightarrow ClO^0 + O_2$$

$$Cl^0 + [O] \rightarrow Cl^0 + O_2$$

Effects of Ozone Depletion:

1. Increased UV Radiation: Thinning of the ozone layer allows more harmful UV radiation to reach the Earth's surface, increasing the risk of skin cancer and cataracts in humans.

2. Impact on Marine Life: Higher UV levels can harm marine ecosystems, affecting phytoplankton, fish larvae, and other aquatic organisms.

3. Agricultural Disruption: Increased UV radiation can reduce crop yields and disrupt ecosystems, impacting food chains.

Green House Effect (Global Warming)

Greenhouse Effect and Global Warming:

Definition: The greenhouse effect is a natural process that warms the Earth's surface. When the sun's energy reaches the Earth, some is absorbed, while the rest is radiated back into space as heat.

Explanation: Greenhouse gases, like carbon dioxide (Co_2), methane (CH_4), and water vapor (H_2O), trap some of this heat, preventing it from escaping the atmosphere. This trapped heat warms the Earth and sustains life as we know it. However, human activities, such as burning fossil fuels and deforestation, have led to an increased concentration of greenhouse gases, enhancing the greenhouse effect and causing global warming.

The process of global warming is primarily driven by the enhanced greenhouse effect, which is caused by an increase in greenhouse gas concentrations in the atmosphere. The most significant greenhouse gas contributing to global warming is carbon dioxide (Co_2). The chemical equation describing the combustion of fossil fuels, which releases CO_2 into the atmosphere, is as follows:

Combustion of Fossil Fuels (e.g., coal, oil, and natural gas):

$$C_6H_{12}O_{6\ \text{(sugar, a simplified representation of hydrocarbons in fossil fuels)}} + 6O_{2\ \text{(oxygen)}} \quad 6CO_{2\ \text{(carbon dioxide)}} + 6H_2O_{\text{(water)}} + \text{energy}$$

This equation represents the burning of fossil fuels, which releases carbon dioxide into the atmosphere. The increasing concentration of CO2, along with other greenhouse gases, traps heat in the Earth's atmosphere, leading to global warming and climate change.

Effects of Global Warming:

1. Rising Temperatures: Global warming leads to higher average temperatures worldwide, resulting in heat waves and extreme weather events.

2. Melting Glaciers and Ice Caps: Warming temperatures cause the melting of glaciers and ice caps, contributing to rising sea levels.

3. Ocean Acidification: Increased CO2 levels in the atmosphere are absorbed by the oceans, leading to ocean acidification, which harms marine life, especially coral reefs.

4. Disruption of Ecosystems: Global warming affects habitats and can lead to changes in the distribution and behavior of species, affecting biodiversity and ecosystem balance.

LINK TO LEARNING

https://phet.colorado.edu/en/simulations/greenhouse-effect

GREENHOUSE EFFECT	GLOBAL WARMING
1. Earth has a special "blanket" of gases around it called the greenhouse effect.	Global warming means the Earth is getting warmer.
2. This blanket keeps the planet warm by trapping some of the sun's heat.	It's happening because people are doing things that add too many greenhouse gases to the air.
3. It's like when you wear a cozy jacket in cold weather to stay warm.	These gases act like too many blankets and make our planet hotter.
4. Greenhouse gases are like the threads in the blanket. They include carbon dioxide and methane.	As it gets hotter, ice at the North and South Poles melts and makes the sea rise.
5. Without this blanket, Earth would be freezing, so the greenhouse effect is good, but too much of it can make our planet too warm.	This can cause problems like floods, big storms, and changes in weather, which is not good for plants, animals, and people.

Water pollution: is the contamination of natural water bodies, like lakes, rivers, and oceans, by harmful substances such as chemicals, sewage, or waste.
Water Suspended Solids: are tiny solid particles, like dirt, silt, and organic matter, that remain suspended in water and can cloud it, making it appear murky or turbid.
Sediments: refer to the larger, heavier particles and debris, such as sand and gravel, that settle at the bottom of a body of water over time, causing the water to become clearer as they settle.
Green chemistry: also known as sustainable chemistry, is a field of chemistry that focuses on designing and developing chemical processes and products in a way that reduces or eliminates the use and generation of hazardous substances.